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REVIEW ARTICLE (INVITED ONLY)

Beyond Water Splitting: Developing Economic Friendly Photoelectrochemical Value-added Reactions for Green Solar-to-Chemical Energy Conversions

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Abstract
Facing climate change and the fossil fuel crisis, the global need for clean energy is more urgent than ever. Converting solar energy to produce green hydrogen fuel through photoelectrochemical (PEC) water splitting is a feasible strategy for utilizing solar energy, aligning with the increasing global demand for clean energy production. However, the energy conversion efficiency and overall economic benefit are severely limited by the slow kinetics of the oxygen evolution reaction (OER) and the low profitability of oxygen production, hindering the commercial utilization of PEC energy conversion technology. In the past few years, studies focusing on replacing the OER with PEC value-added reactions, which can produce high-value products, have received great attention. This not only improves the economic benefits of the general reaction system but also enhances the energy conversion efficiency. In this review, the basic principles of PEC value-added reactions and feedstock-product cost analysis are introduced first. Then, the latest research progress on PEC glycerol oxidation, H_2O_2 production, and ammonia splitting is summarized and discussed in detail. Finally, the status of commercial applications, further prospects, and challenges are discussed.

Keywords Photoelectrochemical · Solar energy · Green hydrogen · H_2O_2 production · Glycerol oxidation · Ammonia Splitting

Introduction

Since the Industrial Revolution, humans have begun to use fossil fuels such as coal, oil, and natural gas on a large scale, leading to a leapfrog development in the productivity of human society [1]. However, with the rapid consumption of fossil fuels, humanity is facing serious challenges. On the one hand, the huge amount of carbon dioxide produced by burning fossil fuels is exacerbating climate change [2]. On the other hand, as demand for energy rapidly grows, many countries are facing an energy shortage crisis. To overcome these challenges, it is necessary to find green and renewable energy sources to replace fossil fuels [3, 4]. Solar energy is one of the most abundant, clean, and sustainable energy sources, which is of great significance to human beings [5]. However, efficiently converting and storing solar energy remains a challenge with current technology. Photovoltaic and photothermal technologies can directly convert solar radiation into electricity with zero carbon emissions and acceptable operating costs [6], but photovoltaic and photothermal power stations must be paired with energy storage systems to convert and store electrical energy that exceeds the grid load into chemical energy, which inevitably leads to energy waste [7].

Photoelectrochemical (PEC) energy conversion technology aligns with these targets and is considered a promising technology for renewable energy production [8]. PEC energy

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conversion is attractive because it directly converts solar energy into stable chemical energy. By absorbing sunlight to generate current that drives electrochemical reactions, inexhaustible solar energy can be converted into stable chemical energy through a clean and efficient pathway [9]. PEC energy conversion was originally proposed by Fujishima and Honda in 1972, showcasing for the first time the application potential for renewable energy production. It produces green hydrogen (H_2) through the water-splitting reaction and converts solar energy into environmentally friendly hydrogen energy [10]. The typical PEC water-splitting process, as shown in Fig. 1a, can be divided into three parts: firstly, the photoanode material is excited to generate electrons and holes under solar illumination; secondly, the holes are transferred to the photoanode surface to complete the oxygen evolution reaction (OER); thirdly, electrons are transferred to the cathode surface to complete the hydrogen evolution reaction [11]. However, since the OER on photoanodes utilizes a four-electron transfer mechanism and requires a high overpotential to drive the reaction, the overall energy conversion efficiency of PEC water splitting is still far from satisfactory for H_2 energy production [12]. Additionally, while using fossil fuels as raw materials for hydrogen fuel production can also yield other chemical products with high economic value [13]. Unfortunately, PEC water splitting only produces hydrogen fuel with the low value-added byproduct of O_2 [14, 15]. This results in PEC energy conversion still being economically uncompetitive with fossil fuels, hindering the commercial application of PEC energy conversion [16]. To address this issue, research into PEC energy conversion has focused on producing value-added products by modifying the PEC anode reactions, which could enable better economic returns and higher energy conversion efficiency [17].

Nowadays, reactions such as PEC carbon dioxide reduction [18, 19], organic upgrading [20, 21], H_2O_2 production [22, 23], and ammonia splitting [24, 25] have been explored as alternatives to PEC water splitting and have achieved considerable research advancements. On the cathode side, the hydrogen evolution reaction (HER) can be replaced with the carbon dioxide reduction reaction (CO_2 RR) [26] or nitrite reduction reaction (NORR) [27, 28]. On the anode side, the oxygen evolution reaction (OER) can also be upgraded to an H_2O_2 production reaction [29], ammonia oxidation reaction (AOR) [30] and series of organic upgrading reaction [31, 32]. These upgraded redox reactions, each with different reaction potentials, are shown in Fig. 1b [33]. Compared to PEC water splitting, these upgraded redox reactions have significant economic advantages. As shown in Fig. 1c, the approximate price of oxygen is around 0.04 USD per kg, while the price of H_2O_2 is around 0.5 USD kg^{-1}, which yields over ten times higher profits [34]. Furthermore, if the anode reaction is replaced with an organic upgrading reaction, such as oxidizing glycerol to 1,3-dihydroxyacetone (DHA), the economic value of the anode reaction can increase thousands of times [35]. Moreover, many organics have lower oxidation potentials for PEC reactions than the standard water oxidation potential, resulting in higher energy conversion efficiency and long-term stability of PEC cells [36].

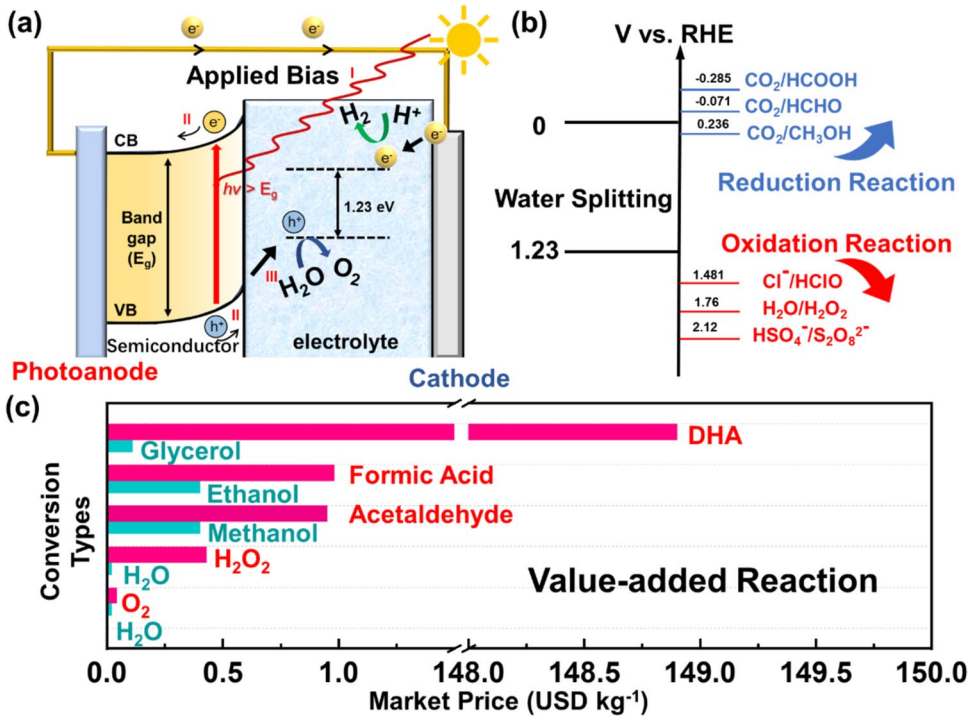

Fig. 1 **a** Schematic illustration of PEC water splitting process. **b** Schematic of series of applied bias of some typical value-added reactions. **c** Illustration of market price for value-added products and cost of feedstocks for some PEC value-added reactions

In this review article, the research status of three different types of anode reactions: PEC glycerol oxidation reaction (GOR), PEC H_2O_2 production, and PEC ammonia oxidation reaction (AOR), is summarized. First, the basic principles and applications of these value-added oxidation reactions will be introduced in detail. Secondly, the latest progress in these upgrading anode reactions, especially the new strategies for synthesizing anode materials and the latest breakthroughs in reaction mechanisms, will be discussed individually. Finally, future challenges that need to be addressed in PEC anode upgrading reactions are explored. We believe that upgrading PEC oxidation reactions will have great potential in solving the difficulties we are facing with the fossil fuel crisis and climate change, thanks to their economic and environmental benefits.

Fundamentals of Anode Reaction Design

Theoretically, the reaction pathway is determined by the thermodynamic properties of the reactants and products [37]. Therefore, each catalytic reaction should take into account its thermodynamic barriers to successfully produce the target product. In the case of PEC reactions, although the difference between the energy band level and the redox potential of the photoanodes can be compensated for by an additional applied bias, a suitable semiconductor should still be carefully selected as the photoanode material in order to minimize the applied bias and reduce power consumption [38]. To overcome the thermodynamic barrier, the energy band gap of the semiconductor should be broader than the sum of the theoretical redox potential and the onset potential of the redesigned anode reaction.

In a traditional PEC water splitting reaction, the redox pair consists of the hydrogen evolution reaction (HER) and the oxygen evolution reaction (OER), with the theoretical redox potential being 1.23 V vs. RHE. The minimum energy required for semiconductors should include activation overpotential and some other minor energy losses [39]. In value-added anode reactions, besides the basic thermodynamic requirements to induce the target redox reaction, the selectivity for value-added chemical products is also a key consideration [40]. On the one hand, upgrading anode reactions and the water oxidation reaction may occur simultaneously on the anode, leading to competitive reactions [41]. On the other hand, there are multiple reaction pathways for GOR [42–44], H_2O_2 production [45–47], and AOR [48, 49], which can lead to series of oxidized products. which can lead to a series of oxidized products. To improve the selectivity of upgrading anode reactions, most recent research on upgrading PEC anode reactions follows these two strategies: (1) modifying the surface chemical structure of anode materials to adjust their thermodynamic properties, and (2) decorating the anode materials with cocatalysts to adjust the reaction kinetics.

The PEC reaction is a heterogeneous catalytic process, and the surface structure is closely related to the reaction pathway [50]. The quasi-Fermi level formed at the semiconductor/electrolyte interface is ensured by the presence of surface states, resulting in minimal impact on the activation energy and rate constants associated with interfacial charge transfer [51]. Different semiconductor surface chemical structures will lead to significant differences in the inherent thermodynamic adsorption characteristics of protons/hydroxide ions. Consequently, even minor changes in the surface structure can significantly affect the adsorption behavior of semiconductors to the electrolyte [52]. This means that the selectivity of the PEC upgrading anode reaction can be improved by adjusting the surface structures of the anode materials. Meanwhile, since most semiconductors exhibit poor surface dynamics compared to metals or metal compounds, decorating with cocatalysts is another promising approach to improve selectivity [53].

Photoelectrochemical Glycerol Oxidation

PEC water splitting is plagued by low energy conversion efficiency and poor economics. In response to the challenges faced by PEC water splitting, researchers have attempted to use organic upgrading reactions to replace water oxidation reactions, thereby improving the overall energy conversion efficiency of PEC energy conversions while obtaining anode products with higher economic value. The catalytic oxidation of glycerol, a surplus byproduct of biodiesel, has received increasing attention due to the popular applications of its oxidation products, such as aldehydes, ketones, and carboxylates. PEC technologies provide a clean and sustainable pathway for glycerol oxidation and upgrading.

Basic Principles of Glycerol Oxidation Reaction

Glycerol is a by-product of biodiesel production, with excess production and low price. Most recently, glycerol's market price is only about 0.66 USD kg^{-1} [54]. Remarkably, by oxidizing glycerol through the dehydrogenation of its secondary hydroxyl groups, 1,3-dihydroxyacetone (DHA), an upgraded product with a high market value of up to 150 USD kg^{-1}, can be produced with no carbon emissions [55]. Additionally, the onset potential of the glycerol oxidation reaction is only 0.4 V vs. the reversible hydrogen electrode (RHE), which is significantly lower than that of the water oxidation reaction (1.23 V vs. RHE), resulting in much more favorable thermodynamic properties [56]. Consequently, when an

aqueous solution of glycerol is used as the electrolyte, the water oxidation reaction on the photoanode surface is easily replaced by the glycerol oxidation reaction, leading to a higher energy conversion efficiency [57–59]. Therefore, replacing the water oxidation reaction with the glycerol oxidation reaction and developing a novel technology for synthesizing DHA coupled with hydrogen production via the PEC glycerol oxidation reaction becomes an attractive research topic. It is a promising strategy to overcome the challenges of low energy conversion efficiency and poor economics in PEC energy conversions.

The research on PEC glycerol oxidation is still in its infancy, with many fundamental issues yet to be resolved. Among these, the most pressing issue is the low selectivity of the glycerol oxidation reaction. The products of PEC GOR are complex, and the reaction process involves a multitude of compounds, including glyceraldehyde, DHA, oleic acid, etc. (Fig. 2) [60]. These substances have similar polarities and are difficult to separate in a co-existing solution, thus limiting the practicality of PEC GOR. From the perspective of chemical reaction principles, the low selectivity of GOR is primarily related to the following factors: (1) Glyceraldehyde is formed by oxidizing the primary hydroxyl group on the glycerol molecule, while DHA is formed by oxidizing the secondary hydroxyl group on the glycerol molecule. Due to the similar reactivity of these two hydroxyl groups, controlling the specific hydroxyl sites involved in the reaction is challenging, which complicates the regulation of selectivity for reaction pathways [61]. (2) The surface adsorption characteristics of the catalyst, the reaction's mass transfer process, and the external environment (such as pH, applied potential, temperature) collectively impact the number of active sites on the catalyst's surface, the degree of oxidation, and the ease of breaking C–C bonds in the glycerol oxidation reaction. These factors can influence the further oxidation of glyceraldehyde or DHA, leading to the formation of substances like oleic acid, taranoic acid, and oxalic acid [62, 63]. In alkaline electrolyte, the glycerol oxidation reaction occurring at the photoanode and photocathode are presented as follows: [64]

$$\text{Anode}: C_3H_8O_3 + 2OH^- \rightarrow C_3H_6O_3 + 2H_2O + 2e^- \quad E = 0.4Vvs.RHE \tag{1}$$

$$\text{Cathode}: 1/2O_2 + 2H_2O + 2e^- \rightarrow 2OH^- \quad E = 0\,Vvs.RHE \tag{2}$$

To make PEC GOR practical for organic synthesis, it is necessary to improve the selectivity of the photoanode reaction, enabling the conversion of glycerol into specific high-purity products (such as DHA and glyceraldehyde) while avoiding the formation of by-products. Currently, research aimed at enhancing the selectivity of PEC GOR primarily follows these two strategies: (1) Adjusting the surface thermodynamic properties of photoanode materials, and manipulating the interaction of photoanode materials with specific hydroxyl groups in glycerol molecules to improve reaction selectivity; (2) Regulating the reaction kinetics of the photoanode materials. This involves promoting the interaction of photogenerated holes with specific surface adsorbed substances (including groups, molecules, or ions), inhibiting the

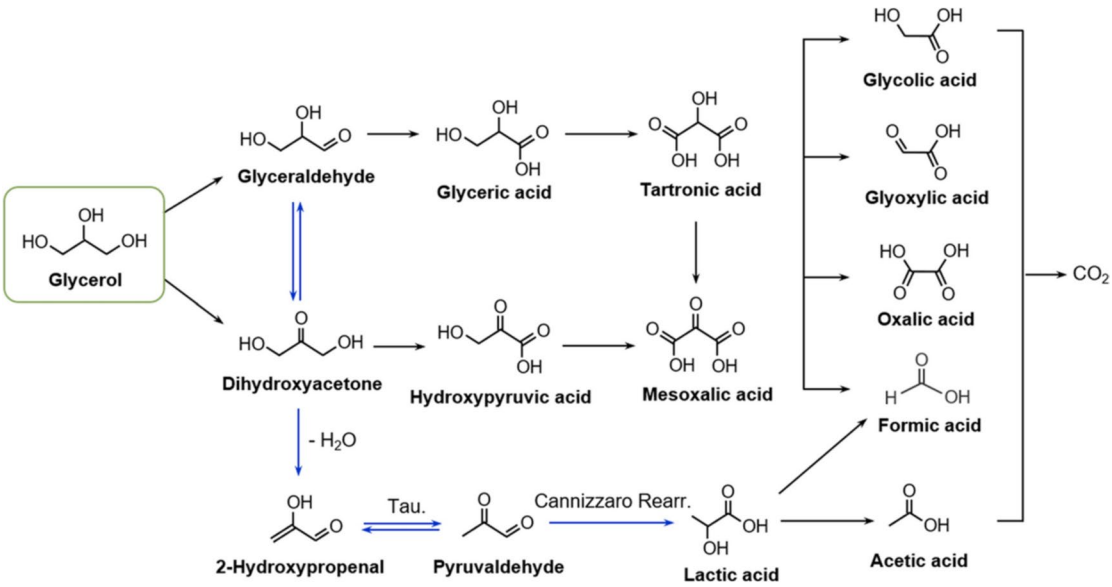

Fig. 2 Possible reaction pathways and main products derived from the catalytic oxidation of glycerol. Adapted from ref [60]. Copyright 2023 Springer

occurrence of side reactions, and reducing the desorption activation energy of the target product. As a result, both the selectivity and conversion rate of the target product can be significantly improved.

Transition Metal Oxides for PEC GOR

In early studies, researchers attempted to use Au, Pt, and other precious metal materials as catalysts for the electrocatalytic and photocatalytic oxidation of glycerol. Although noble metal catalysts showed activity in producing glyceraldehyde and DHA, they were not suitable for PEC glycerol oxidation due to their high cost and tendency to over-oxidize glyceraldehyde and DHA. In recent years, with the increasing research focus on PEC GOR, researchers have begun exploring the use of TiO_2 [65], WO_3, [66–68] α-Fe_2O_3, [69–71] $BiVO_4$ [72–75] and other transition metal oxides (TMOs) [76–78] as photoanode materials in this area. By reasonably modifying the TMOs, efforts have been made to improve the selectivity for DHA in GOR from various perspectives, such as enhancing adsorption properties, improving electron conduction properties, adjusting the energy band structure, and increasing light absorption.

In PEC reactions, holes are transferred to the surface of the photoanode to undergo an oxidation reaction, while electrons are transferred to the surface of the cathode to complete the reduction reaction [79]. Therefore, an n-type semiconductor is usually used as the photoanode, and a p-type semiconductor is used as the cathode. To achieve superior PEC performance, the semiconductors used in photoelectrodes should possess a sufficiently wide band gap, high catalytic activity, and chemical stability [80]. Transition metal oxides represented by TiO_2, WO_3, α-Fe_2O_3 and $BiVO_4$ exhibit typical n-type semiconductor characteristics and are cost-effective, thus fully meeting the basic requirements [81, 82].

TiO_2 is the initial TMOs employed for PEC GOR. The team led by Mayank Misra at the University of Nevada pioneered the application of TiO_2 nanotube arrays as the material for photoanodes in conducting PEC GOR experiments in a 1 M KOH solution under ultraviolet (UV) light [83]. They were the first to establish that GOR occurs more favorably, both thermodynamically and kinetically, than water oxidation reactions at the photoanode. Hematite (α-Fe_2O_3), with its 2.2 eV band gap, can absorb visible light, though its efficacy in water oxidation is compromised by rapid electron/hole recombination. In 2021, a breakthrough was made by Elton Sitta's team, who enhanced the performance of α-Fe_2O_3 photoanodes by incorporating glycerol into the KOH solution. This addition significantly increased the electrooxidation current density and substantially reduced the onset potential. [84] The findings from Sitta's group indicated that with glycerol, the oxidation reactions predominantly occur through glycerol oxidation rather than evolving oxygen. The dependence of photocurrent on temperature was linked to temperature's effect on various interface processes involving both the electrode and the solution, specifically: electron/hole pairs generation and mobility within α-Fe_2O_3, the coverage and reactivity of intermediate surface species, and the diffusion of glycerol from the bulk solution to the interface. However, it was also discovered that α-Fe_2O_3 could entirely break down the C–C chains in glycerol, leading to its over-oxidation into a mix of C2 compounds, including glycerate, lactate, and glycolate [84].

$BiVO_4$ serves as an excellent semiconductor material for photoanodes, offering an appropriate band structure and robust resistance to photo-corrosion. Lu–Wei Huang et al. first developed a PEC cell based on $BiVO_4$ photoanode in 2019 (Fig. 3a), using photoanode glycerol oxidation reaction to assist hydrogen production in a mixed electrolyte solution composed of 0.1 M $Na_2B_4O_7$ (pH = 9.4) and 0.1 M glycerol under 1 sun illumination. The research highlighted how glycerol functions as a hole scavenger in the oxidation–reduction process, effectively preventing the recombination of surface electron holes on $BiVO_4$. This process significantly enhanced the efficiency of photocatalytic hydrogen generation. Furthermore, by oxidizing glycerol, the PEC cells were capable of producing both DHA and formic acid, achieving a product selectivity rate of approximately 15% for DHA (Fig. 3b,3c) [85].

WO_3 stands out as another prominent semiconductor frequently explored in PEC energy conversions. It has been found that alkaline environments enhance the cleavage of C–C bonds during glycerol oxidation, leading to the products being excessively oxidized into formic acid and CO_2. [86] Thereby, if the PEC GOR takes place in acid or neutral condition, it can not only suppress the over-oxidation, but also facilitate the cathode hydrogen evolution reaction [86]. Thus, conducting PEC GOR in acidic or neutral settings can mitigate this over-oxidation while simultaneously promoting the hydrogen evolution reaction at the cathode. Owing to its high chemical stability in acidic and neutral pH conditions, WO_3 is seen as having significant potential in PEC glycerol oxidation applications. However, research on WO_3 that is used in PEC GOR remains sparse. As shown in Fig. 3d–g, Jie Yu and colleagues examined the PEC behavior of monoclinic WO_3 in oxidizing glycerol to higher-value chemicals, such as DHA and glyceraldehyde, showing promising outcomes. [87] This specific form of WO_3 was observed to outperform commercially available WO_3 catalysts in terms of light absorption, PEC activity for glycerol oxidation, current density, and the rate of product enhancement [87]. The study revealed that light exposure and the use of an external bias initiate glycerol oxidation on

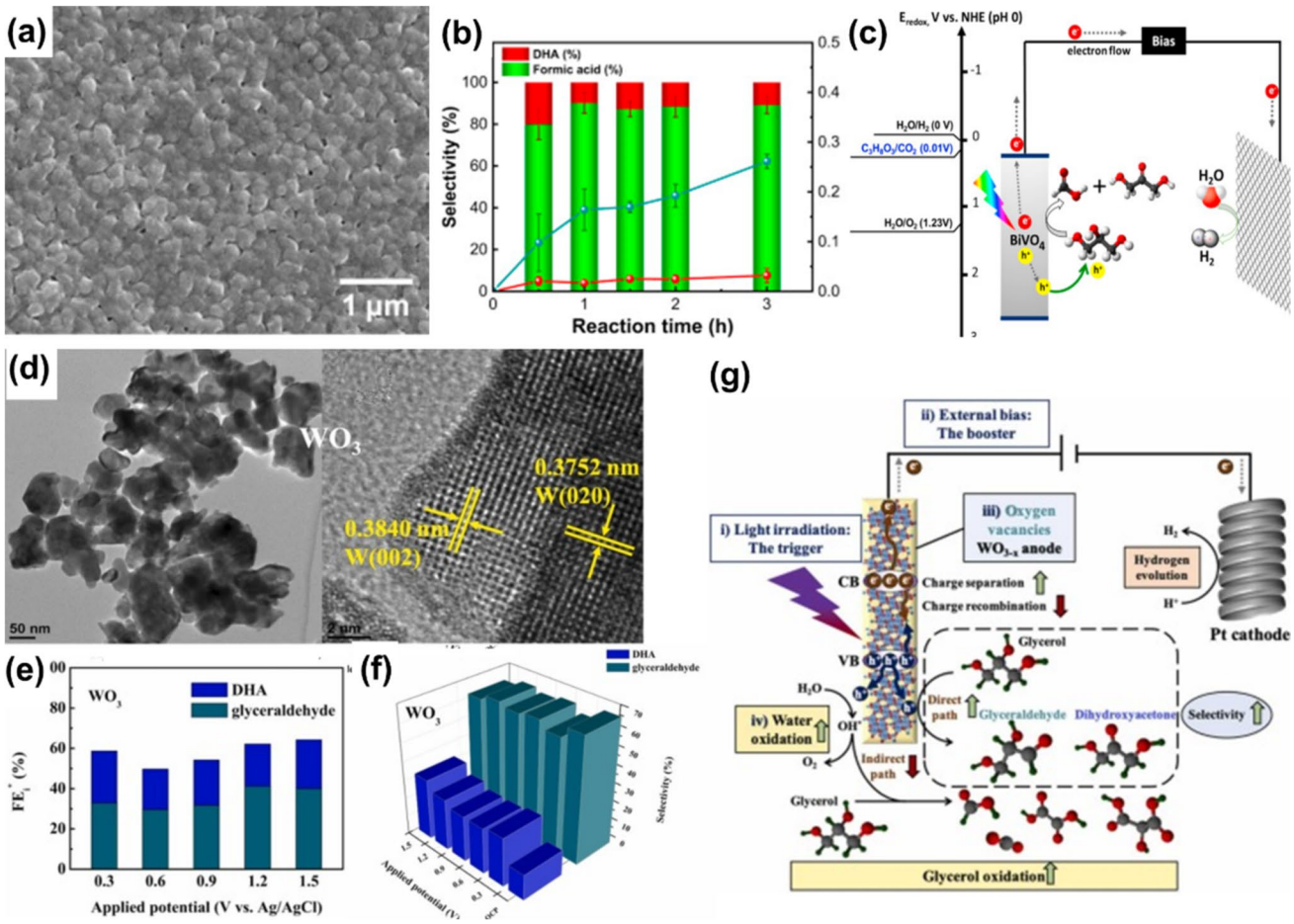

Fig. 3 **a** Morphology of $BiVO_4$ photoanode for PEC glycerol oxidation and **b** products selectivity of $BiVO_4$ photoanode. **c** Mechanism illustration of $BiVO_4$ photoanode in PEC GOR. Adapted from ref [85]. Copyright 2019 Elsevier. **d** Morphology of WO_3 photoanode for PEC glycerol oxidation and **e** products selectivity and **f** Faradic efficiency of WO_3 photoanode. **g** Mechanism illustration of WO_3 photoanode in PEC GOR. Adapted from ref. Adapted from ref. [87]. Copyright 2023 Elsevier

monoclinic WO_3, which is further enhanced by the presence of oxygen vacancies. These vacancies improve the semiconductor's light responsiveness, charge carrier separation, and electrical conductivity. The enhanced charge carrier separation effectively prevents recombination, making more holes available for both water and glycerol oxidation reactions. Monoclinic WO_3 showed a high and stable selectivity for liquid products, particularly glyceraldehyde and DHA, with selectivity rates between 87 and 95% within a potential range of 0–1.5 V vs. Ag/AgCl.

Semiconductors like WO_3 and $BiVO_4$ show promising capabilities for enhancing glycerol conversion in PEC processes. However, there's a need to boost their catalytic efficiency and the selectivity for particular products during glycerol upgrading. Thus, it becomes essential to tailor the properties of TMOs to achieve desired outcomes in glycerol enhancement performance.

Surface Manipulated TMOs for PEC GOR

While TMOs-based photoanodes have exhibited promising performance in a range of PEC activities, including water and glycerol oxidation, they often encounter several limitations such as short electron/hole diffusion length, rapid electron/hole recombination, sluggish surface charge transfer, limited conductivity, slow reaction rates, and compromised chemical durability, all of which hinder their broader application [88, 89]. As is mentioned above, efforts to enhance the glycerol upgrading efficiency of TMOs have primarily concentrated on modifying their surface thermodynamic characteristics and optimizing their reaction kinetics. Consequently, various approaches have been explored to improve the performance of TMO

semiconductors in glycerol upgrading, including crystal structure modification and the addition of cocatalysts.

Altering the primary exposed crystal face of TMOs is an effective method for tuning the selectivity towards specific products in glycerol oxidation reactions [90]. Truong-Giang Vo et al. developed two variants of $BiVO_4$, each characterized by distinct crystal faces, predominantly {010} and {121} crystal facets, achieved by modifying the pH of the precursor solution (Fig. 4a) [91]. Because {010}-$BiVO_4$ exhibits a more pronounced adsorption effect on the secondary hydroxyl groups in glycerol molecules (Fig. 4b), $BiVO_4$ photoanodes with a predominance of the {010} crystal face has a specific affinity for these secondary hydroxyl groups in glycerol. Additionally, {010}-$BiVO_4$ benefits from a higher carrier density, enhancing the electron conductivity of the photoanode and facilitating the charge transfer process. This leads to increased selectivity for DHA, (Fig. 4c–e) [91]. However, the DHA product selectivity of the modified $BiVO_4$ photoanode is only about 50%, indicating a need for further advancements in the photoanode material [91].

The predominant exposure facets of WO_3 also demonstrates close relationship to their glycerol upgrading properties. Jie Ouyang et al. synthesized a series of monoclinic WO_3 materials, each with different leading exposure facets, and systematically investigated their properties in glycerol oxidation reactions through theoretical studies [92]. Quasi-cubic WO_3 crystals exhibit comparable exposure of the {002}, {020}, and {200} facets. n contrast, sheetlike WO3 nanoplates restrict growth in two directions, favoring the exposure of one among these three facets. Density functional theory (DFT) calculations reveal a hierarchy in surface energy: {002}-WO_3 (1.56 J m^{-2}) > {020}-WO_3 (1.54 J m^{-2}) > {200}-WO_3 (1.43 J m^{-2}), thus indicating that {200}-WO_3 is the most stable [92].

While some researchers have aimed to enhance the selectivity of glycerol oxidation reactions on photoanodes by modifying the dominant exposed crystal facet of the TMO photoanode materials, achieving high DHA selectivity remains undocumented. Hence, optimizing the primary exposed crystal facet of WO_3 photoanode materials to increase DHA selectivity in glycerol oxidation remains a significant area for research. Commonly investigated WO_3 crystal facets are of low indices, yet high-index facets, which exhibit a higher degree of atomic unsaturation, show elevated catalytic activity. For glycerol oxidation on photoanodes, the current density of {202}-WO_3 attains 3.52 mA cm^{-2} at 1.23 V vs. RHE, more than double that of the {200}-WO_3 photoanode. The glyceraldehyde selectivity for {202}-WO_3 is remarkably high at 80%, attributed to its superior thermodynamic properties in PEC GOR [93]. The glycerol's terminal hydroxyl group binds more readily than the middle hydroxyl group, with the adsorption

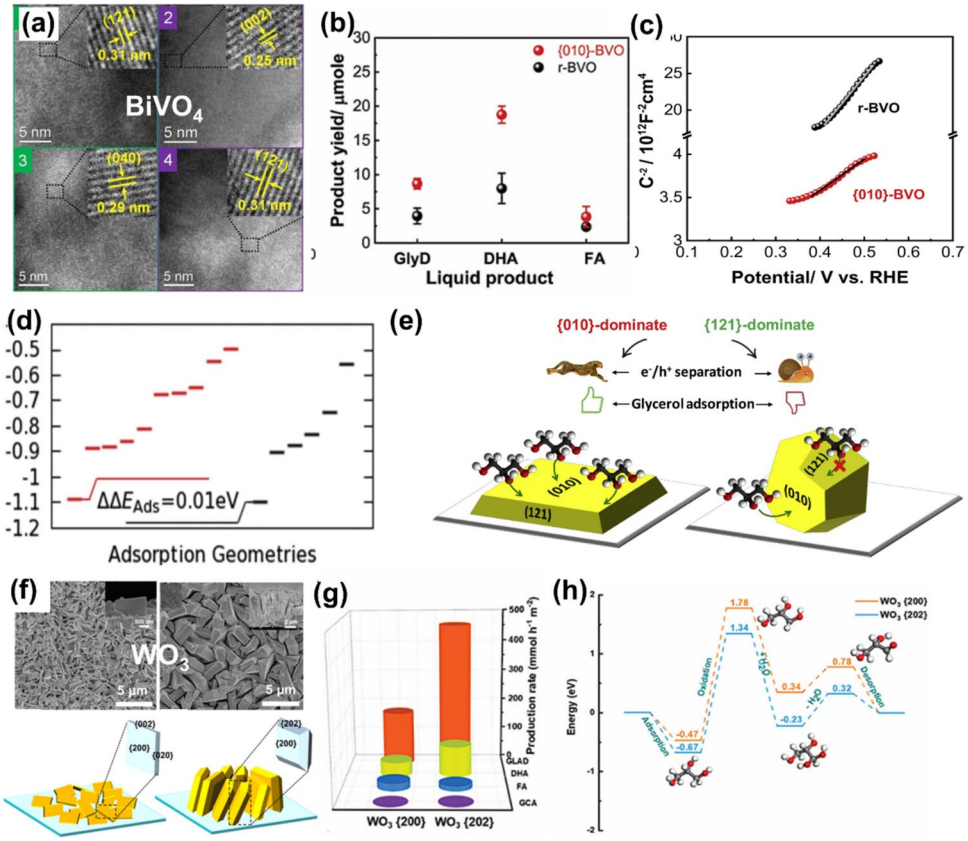

Fig. 4 **a** Crystalline facets characteristic of series $BiVO_4$, and **b** production yield, **c** carriers migration behaviors, **d** thermodynamic characteristics of $BiVO_4$ photoanodes with different facet. e Illustration of the effect of kinetics and thermodynamic factors to PEC GOR. Adapted from [91]. Copyright 2020 Elsevier. **e–h** Crystalline structure, production yield and thermodynamic characteristics of WO_3 photoanodes with different facets for PEC GOR. Adapted from [92].Copyright 2022 Elsevier

energy on the {202} facet being lower than on the {200} facet (Fig. 4f-h) [93]. The {200} planes have a flat atomic surface arrangement, whereas the {202} planes feature a zig-zag structure, making the tungsten and oxygen atoms more accessible for interaction with glycerol's oxygen and hydrogen. The adsorption of glycerol molecules on tungsten oxide elongates the H–O bond in the terminal hydroxyl groups, indicating glycerol activation. The critical step for both photoanodes involves the uptake of a hole and the removal of an alpha hydrogen from the C–H bond. The {202} facet shows a lower thermodynamic hurdle for this step than the {200} facet, aligning with its enhanced PEC activity. The ensuing radical undergoes a spontaneous hydration reaction in water, forming a gem diol species. This gem diol is transient and dehydrates, resulting in glyceraldehyde formation, which accounts for the elevated selectivity towards glyceraldehyde [93].

Cocatalyst decoration is another well-known strategy to adjust the surface properties of TMOs semiconductors. The surface reaction kinetics of photoanodes in standard anode reactions are often slow, owing to the involvement of complex multi-electron transfer processes. This sluggishness not only affects the reactions at the cathode but also lowers the overall efficiency of energy conversion in PEC reactions. Decorating with a cocatalyst can address this problem by enhancing the reaction kinetics at the anode. Nickel-based materials, such as nickel phosphide, nickel phosphate, and nickel oxyhydroxide, are frequently utilized as cocatalysts for this purpose [94–97].

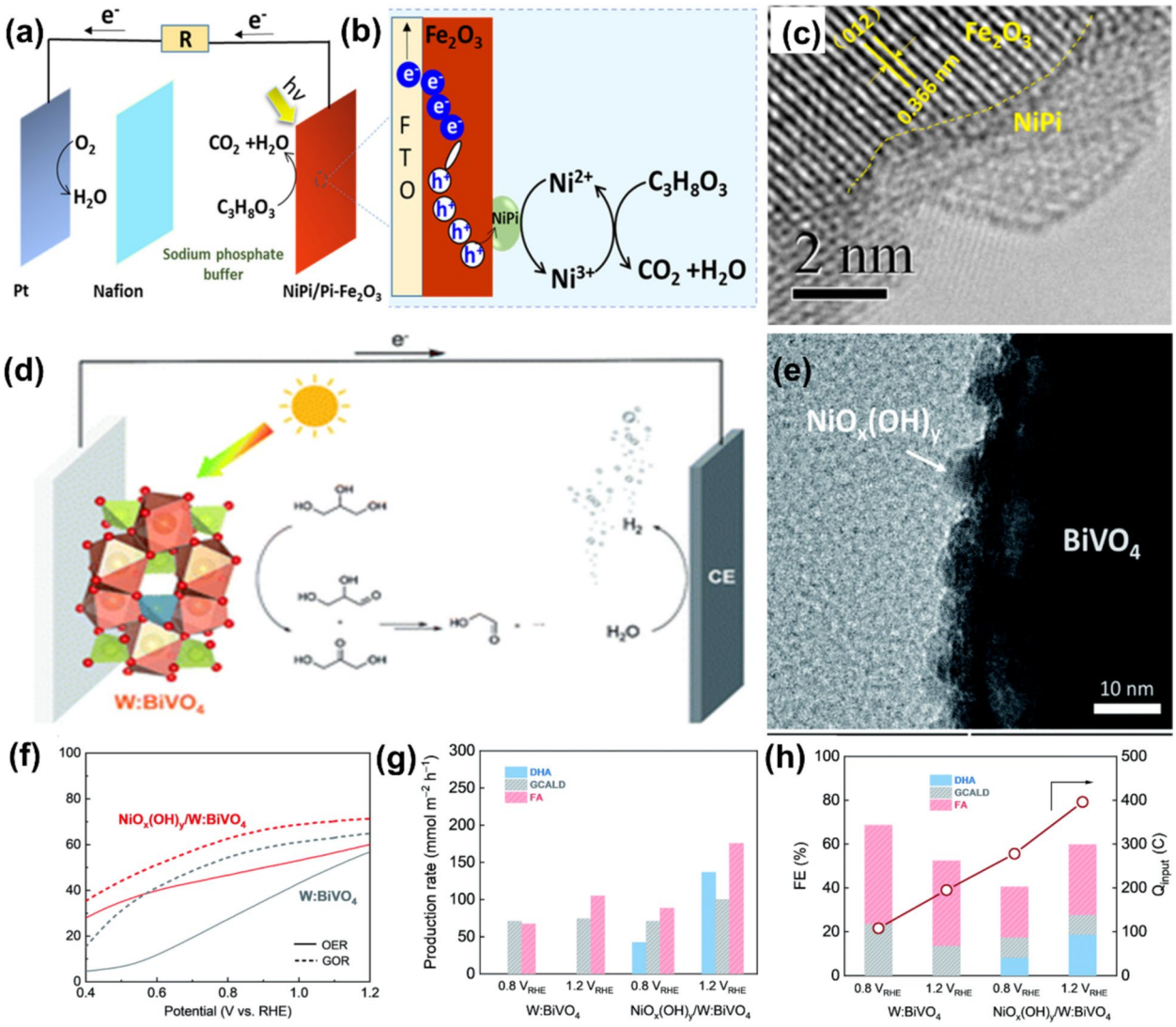

Fig. 5 **a-c** Illustration and morphology for NiPi/Pi-Fe_2O_3 for PEC glycerol-based fuel cell. Adapted from [98]. Copyright 2017 Elsevier. **d-h** Mechanism illustration, performance, and characteristics of $NiO_x(OH)_y$/W:$BiVO_4$ photoanode for PEC GOR. Adapted from [99]. Copyright 2021 Royel Scoiety of Chemistry

Ruifeng Chong et al. decorated nickel phosphate (NiPi) on the surface of α-Fe_2O_3 photoanode to construct a glycerol fuel cell that can operate at neutral pH [98]. As shown in Fig. 5a-c, NiPi serves a dual function: it not only prevents the recombination of electron/hole pairs, enhancing the reaction efficiency, but also boosts the kinetics of glycerol oxidation via the reversible [98] This results in significantly improved charge transfer at the semiconductor/liquid boundary and increased hole injection into the electrolyte, with NiPi acting both as a cocatalyst for capturing photogenerated holes from $\alpha-Fe_2O_3$ and as a catalyst providing active sites for glycerol oxidation [98]. Yi-Hsuan Wu et al. modified an amorphous nickel-based cocatalyst (NiOx(OH)y) to W:$BiVO_4$ photoanode and tested effect of $NiO_x(OH)_y$/W:$BiVO_4$ photoanode on photocatalytic glycerol oxidation material (Fig. 5d, e) [99]. In an alkaline electrolyte, the kinetics of glycerol oxidation on the photoanode were substantially improved through facilitated hole transfer from the electrode surface to glycerol, prompted by the nickel-based cocatalyst (Fig. 5f). This modification also reduced the onset potential of the photoanode reaction by about 50 mV compared to the unaltered W:$BiVO_4$ photoanode, and notably increased the DHA yield (Fig. 5g) [99]. Moreover, this research marked the first detection of glycolaldehyde (GCALD) as a by-product of PEC glycerol oxidation. It's hypothesized that GCALD forms from isomeric DHA/GALD species via C–C bond cleavage, releasing one equivalent of formic acid (FA). Further cleavage of the C–C bond in GCALD is suggested to produce an additional equivalent of FA and formaldehyde (FMALD). (Fig. 5h) The identification of hydrated GCALD through HPLC and NMR spectroscopy confirms that PEC GOR inevitably leads to the formation of multiple products.

Summary of PEC GOR

A summary table is given for recent progress of PEC glycerol oxidation as Table 1. While recent studies have shown that selectively creating upgraded products from glycerol is possible, the issue of low selectivity for specific products has not been fully resolved. There is still a need for more efficient cocatalysts to enhance the purity of these value-added products.

The process of PEC glycerol oxidation is highly intricate, yielding a variety of products such as DHA, glyceraldehyde, oleic acid, and others. However, the selectivity for a specific end product, particularly for valuable compounds like DHA or glyceraldehyde, remains insufficient for commercial purposes. Currently, PEC glycerol oxidation technology is in a nascent phase of exploration, highlighting the need for a comprehensive investigation into its kinetic mechanisms and a clear understanding of how various oxidation products are formed. Additionally, uncovering the kinetic mechanisms that allow for the manipulation of oxidation products in PEC GOR is crucial.

At presents, the industrial-scale synthesis of organic substances such as DHA and glyceraldehyde via PEC GOR has not yet been achieved. In addition to further improving the selectivity of the PEC GOR reaction, in-depth research should also be conducted on the design of the reactor.

Table 1 Progress of value-added upgrading production via PEC glycerol oxidation

Material	Main Product	Electrolyte	Current Density (mA cm^{-2})	Selectivity (%)	Applied Bias (V vs. RHE)	Ref
TiO_2	–	1 M KOH + 5vol % glycerol	3.3	–	1.23	[83]
WO_3/TiO_2	Glyceraldehyde	0.5 M Na_2SO_4 + 0.1 M glycerol + 0.1 M boratebuffer	2.8	62	1.2	[143]
α-Fe_2O_3	Glycolate	0.1 M KOH + 1.0 M glycerol	0.75	58	1.23	[84]
$BiVO_4$	Formic Acid	0.1 M $Na_2B_4O_7$ + 2.0 M glycerol	1.28	85	1.23	[85]
$BiVO_4$	DHA	0.1 M $Na_2B_4O_7$ + 0.1 M glycerol	1.4	60	1.1	[91]
$BiVO_4$	DHA	0.5 M Na_2SO_4 + 0.1 M glycerol	3.7	51	1.2	[144]
$BiVO_4$	DHA	0.5 M Na_2SO_4 + 0.1 M glycerol	6.04	53.7	1.23	[145]
$NiO_x(OH)_y$/ W:$BiVO_4$	Formic Acid	0.5 M Na_2SO_4 + 0.1 M glycerol	3.3	37	1.2	[99]
$BiVO_4$/NiCo-LDH-Act	DHA	0.5 M Na_2SO_4 + 0.6 M glycerol	5.5	41.93	1.4	[146]
Ta_2N_5/CoNiFe-LDH	Formic Acid	1 M NaOH + 10 vol% glycerol	3.59	100	1.23	[147]
Bi_2O_3/TiO_2	DHA	0.5 M Na_2SO_4 + 0.1 M glycerol	2.3	75.4	1.0	[148]
TiO_2(N–C)/ CPB/TiO_2	DHA	0.1 M NaOH + 10 vol% glycerol	4.5	51	1.2	[149]
WO_3	Glyceraldehyde	0.5 M Na_2SO_4 + 0.1 M glycerol	0.3	65	1.23	[93]
WO_3	Glyceraldehyde	0.1 M Na_2SO_4 + 0.1 M glycerol	3.5	80	1.23	[92]

Scaling up the reactor to an industrial operating unit requires fabrication of large-area electrodes that do not require membranes to reduce operating costs, which can be achieved when the same electrolyte is used in both cathode and anode compartments. The integration of PV-PECs for glycerol oxidation remains to be explored and requires optimization of the reactor design to efficiently produce the best potential.

Photoelectrochemical H_2O_2 Production

Beside of PEC GOR, researchers have also tried to use PEC reactions to catalyze water splitting to produce H_2O_2. This section focuses on research progress of PEC H_2O_2 production which occurring at the anode that produces valuable H_2O_2 from water splitting.

Fossil fuel shortage and environmental pollution are two serious problems that human society are facing at this century. H_2O_2 has a high energy density of 3.5 MJ kg^{-1}a, which is almost as the energy density of H_2. During the past centuries, H_2O_2 has been widely used in papermaking, chemical industry, food industry, environmental protection and other fields. It is also one of the few chemical substances that can attract research interest in both energy and environmental fields [100]. Currently, the industrial production of H_2O_2 is mainly through the "anthraquinone autoxidation method", which requires the use of expensive Pd catalysts and relies on large bubbling towers as oxidation reactors [101]. Indeed, the Pd catalyst used has drawbacks, including poor corrosion resistance and poor long-term stability, resulting in expensive manufacturing and operational costs [102]. In addition, high concentrations of H_2O_2 also pose great safety risks during transportation. Due to the cumulative effect of the above factors, the cost of industrial H_2O_2 production is still high [102].

PEC H_2O_2 production can utilize water as a feedstock. With the easy acquisition of feedstock, low cost, and coupled H_2 production, it is considered an ideal method for producing green H_2O_2. [103] As is shown in Fig. 6a, a typical model of PEC H_2O_2 production involves oxidizing H_2O to H_2O_2, instead of O_2, at the anode via the two-electron water oxidation reaction ($2e^-$ WOR) [104]. However, this reaction competes with the four-electron oxygen evolution reaction ($4e^-$ WOR), which has a lower thermodynamic energy barrier, and H_2O_2 is easily decomposed by oxidation, making it challenging to achieve high product selectivity, as shown in Eqs. (3) and (4): [105]

$$\begin{aligned} &4e^- WOR: 2\,H_2O + 4e^-/h^+ \rightarrow O_2 + 2H_2 \\ &E = 1.23\,V\,vs.\,RHE \end{aligned} \tag{3}$$

$$\begin{aligned} &2e^- WOR: 2\,H_2O + 2e^-/h_2^+ \rightarrow H_2\,O_2 + H_2 \\ &E = 1.76\,V\,vs.\,RHE \end{aligned} \tag{4}$$

As early as 1976, M.S. Wrighton et al. detected the formation of H_2O_2 in water oxidation during the reaction of hydrogen production from water decomposition by a $SrTiO_3$ photoelectrode, but no breakthroughs have been made in subsequent studies [106]. Therefore, how to improve the selectivity of H_2O_2 precipitation by water oxidation in photoanode has become one of the problems that have puzzled the scientific community for a long time.

Basic Principles of PEC H_2O_2 Production

In 2015, theoretical chemist Professor Nørskov at Stanford University analyzed the critical rate steps in forming H_2O_2 and O_2 products based on the thermodynamic principles of water oxidation intermediates and the relationship between the energy ratios of the products. This analysis led to the prediction of several catalysts capable of selectively oxidizing water to H_2O_2. In the oxygen evolution reaction, semiconductors with strong OH adsorption energy transform OH∗ to O∗ and OOH∗, leading to the release of O_2 through a complete four-electron transfer. Conversely, semiconductors with weaker OH∗ free energy are more inclined to produce H_2O_2 via a two-electron transfer. The free energy of H_2O_2 is 3.52 eV, double its theoretical equilibrium potential of 1.76 eV. Theoretically, the ΔG_{O*} of catalyst surface should be higher than 3.52 eV. For various oxide systems, scaling relations have been established between reaction intermediates; $\Delta G_{O*} = 2\Delta G_{OH*} + 0.28$ for metal-oxide surfaces. Considering the energy ratio of O∗ and OH∗, the minimum free energy of OH∗ should exceed 1.6 eV. The maximum ΔG_{OH*} is determined by the free energy required for OH radical formation in the system, which is $\Delta G_{OH*} > 2.38$ eV. Thus, thermodynamic evaluation and product energy ratio analysis suggest that catalysts for selective oxidation of H_2O to H_2O_2 should have a ΔGOH∗ ranging between 1.6 eV and 2.4 eV [107]. As shown in Fig. 6b, based on the thermodynamic parameters ΔG_{OH*} and ΔG_{O*}, semiconductors such as TiO_2, $BiVO_4$, and WO_3 fulfill the thermodynamic prerequisites for selectively generating H_2O_2 through photoelectrochemical [108]

It is known that in the presence of multiple competing reactions during a target reaction process, the distribution of products can be influenced by leveraging the differences in the formation rates of various products, without altering the reaction's thermodynamic properties. The primary goal of controlling the formation rate of products, also referred to as kinetic regulation, aims to extract the target

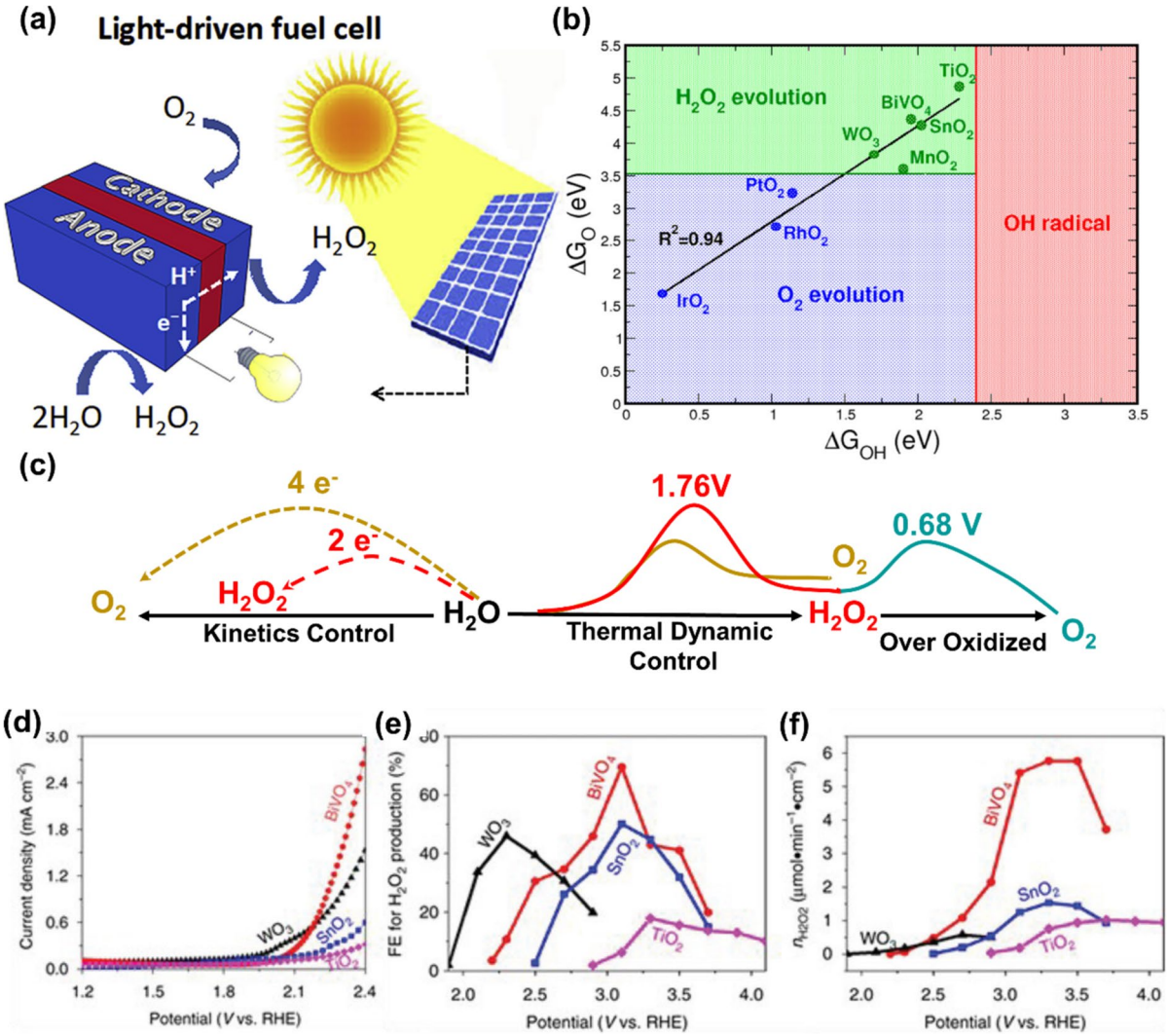

Fig. 6 **a** Illustration of solar-driven PEC cell for H_2O_2 production via 2e-WOR. Adapted from [104] Copyright 2021 Cell. **b** Activity volcano plot of series TMOs for H_2O_2 production. Adapted from [107]. Copyright 2017 Springer. **c** Illustration of regulation strategies for highly-efficiency PEC H_2O_2 production. **d-f** Current density, Faraday efficiency and production yield of H_2O_2 for series transition metal oxides. Adapted from [107]. Copyright 2017 Springer

product from the catalyst mass before the reaction achieves equilibrium. For instance, in the electrocatalytic process of oxygen evolution, a high concentration of O_2 on the catalyst surface can lead to slow desorption kinetics for O_2, significantly diminishing the reaction rate [109, 110]. In the aqueous electrocatalytic reduction of CO_2 and O_2, hydrophobic catalysts are typically employed to boost the concentration of gas molecules within the surface reaction zone, thereby enhancing the reaction kinetics. [111] Within the thermodynamic range, optimizing the kinetics of the target product's formation is a viable strategy [112]. Therefore, as shown in Fig. 6c, a guiding strategy for current studies on PEC H_2O_2 production is to restrict the diffusion of O_2 at the material level and regulate the charge transfer rate at the energy level, which is expected to avoid inherent thermodynamic barriers and improve the reaction kinetics of water oxidation to H_2O_2, so as to achieve high selectivity and stability of H_2O_2 production via PEC water oxidation.

PEC H_2O_2 production reaction is mainly achieved through the indirect way of HCO_3^- self-oxidation to percarbonate (HCO_4^- and $C_2O_6^{2-}$), which then deoxidizes water to form H_2O_2. The mechanism of this method is similar to that of anthraquinone autooxidation method used in industry, so it shows better product selectivity. In recent years, with the assistance of HCO_3^-, many metal oxide photoanodes, such as TiO_2 [107], Bi_2WO_6 [113], ZnO [114], $BiVO_4$ [115, 116], etc., have shown good activity of water oxidation to produce H_2O_2.

Nørskov's research group identified trends in the activity of water oxidation for H_2O_2 production across four distinct metal oxides, namely WO_3, SnO_2, TiO_2, and $BiVO_4$, utilizing both DFT simulations and empirical data [107]. They discovered that among the oxides tested, $BiVO_4$ exhibited the best performance for H_2O_2 production, in terms of both current density and faradaic efficiency. As shown in Fig. 6e-g, the order of both the measured current onset potential and the onset potential for H_2O_2 production is $WO_3 < BiVO_4 < SnO_2 < TiO_2$, aligning with the theoretical predictions in Fig. 6b. [108] Regarding the faradaic efficiency and the H_2O_2 yield from the oxides under different applied biases, $BiVO_4$ exhibited the greatest FE, reaching 70%, and the highest production rate of approximately 5.9 μm min^{-1} cm^{-2}. All the TMOs studied displayed a similar FE trend with increasing applied bias: FE initially rises to a peak and then declines. Each TMO has an optimal potential range for producing H_2O_2, stemming from the competitive dynamics between two-electron and four-electron oxidation reactions, as predicted theoretically. $BiVO_4$ stands out among the oxides tested, attaining the peak FE of 70% at 3.1 V vs RHE. Moreover, $BiVO_4$ demonstrated the highest rate of H_2O_2 generation per geometric area of the electrode, a result of its high FE coupled with substantial current density. The exceptional H_2O_2 generation performance of $BiVO_4$ is likely due to its favorable optical band gap (2.4 eV) and the low position of its valence band bottom (2.83 eV) [108].

Cocatalyst Decorating for PECH_2O_2 Production

In order to achieve higher H_2O_2 production performance, a series of methods to improve solar energy utilization have been developed, such as elemental doping [117], crystal facet regulation [118], surface passivation [105], and heterogeneous structure construction [119]. However, the PEC H_2O_2 production is a reversible reaction, which is restricted by chemical equilibrium, and the concentration of H_2O_2 produced by this way is still limited in ppm level, which is difficult to meet the needs of industrial applications.[120] Therefore, the current mainstream strategy for PEC H_2O_2 production is to regulate the kinetic characteristics of semiconductors in H_2O_2 reactions. By limiting the diffusion of O_2 from the mass transfer view and regulating the charge transport from the energy view, inherent thermodynamic barriers are avoided. Finally, the H_2O_2 production via water oxidation is promoted by regulating reaction kinetic factors.

Decorating with cocatalysts remains a fundamental approach for modulating the reaction kinetics of the PEC H_2O_2 production process. Kan Zhang et al. applied a SnO_{2-x} overlayer onto a $BiVO_4$ photoanode to balance the competition between the OER and the H_2O_2 evolution reaction on the photoanode (Fig. 7a) [105]. The presence of the SnO_{2-x} cocatalyst overlayer significantly inhibited the OER at the photoanode side. Introducing the SnO_{2-x} overlayer reduced the quasi-hole Fermi energy at the bulk/electrolyte interface and the thermodynamic Gibbs free energy difference between the competitive two-electron and four-electron reactions, thus enhancing the transfer of holes from $BiVO_4$ to the SnO_{2-x} layer. This adjustment was kinetically advantageous for H_2O_2 evolution, achieving high selectivity through diminished band bending. The generation of H_2O_2 was facilitated by the creation of -OH groups via one-electron water oxidation reactions. (Fig. 7b). [105] Beyond promoting H_2O_2 evolution from the PEC water oxidation reaction through H_2O oxidation, the $SnO_{2-x}/BiVO_4$ photoanode crafted also plays a role in preventing the over-oxidation of H_2O_2 into O_2, contributing to sustained H_2O_2 production over an extended period.

By appropriately combining cocatalysts with different properties, the slow water oxidation kinetics can be improved while promoting downstream gas separation. As shown in Fig. 7c**,** Yang An et al. employing a $CaSnO_3$/$SrTiO_3$ hybrid cocatalyst overlayer on $BiVO_4$ (CSO/STO/BVO). [121] The interface states that arise between BVO and CSO can act as centers for recombination, leading to significant interface recombination. This results in a reduction of surface-reaching holes available for the two-electron transfer process. To mitigate this detrimental effect, a hole extraction layer is commonly employed, where the impetus for charge separation and movement becomes essential. STO, whose charge-transport characteristics can be modified through ferroelectric polarization, was inserted between BVO and CSO. The STO/BVO interface, with its advantageous band alignment for charge movement and the internal electric field created by polarized ferroelectric domains, contributed to enhanced PEC efficiency in the CSO/STO/BVO system. This setup showcased a characteristic trait of ferroelectric materials. Consequently, with positive polarization at the photoanode, a negative ferroelectric charge accumulates at the BVO/STO interface, intensifying BVO's band bending, which in turn improves charge separation and transport (Fig. 7d). On the other hand, negative polarization reverses the direction of the electric field, raising the energy barrier for hole migration, thereby reducing the number of holes that can traverse the BVO/STO interface for the water oxidation reaction at the surface (Fig. 7e)**.** The photoanode, through the incorporation of the STO/CSO hybrid cocatalyst, achieves elevated PEC activity and faradaic efficiency for H_2O_2 production (Fig. 7f–g). Furthermore, the deployment of integrated array panel reactors, which are amenable to scalable manufacturing techniques, has demonstrated high energy conversion efficacy on a commercial scale [121].

Beside $BiVO_4$, TiO_2 is also a promising candidate for PEC H_2O_2 production. Jaekyum Kim et.al. used

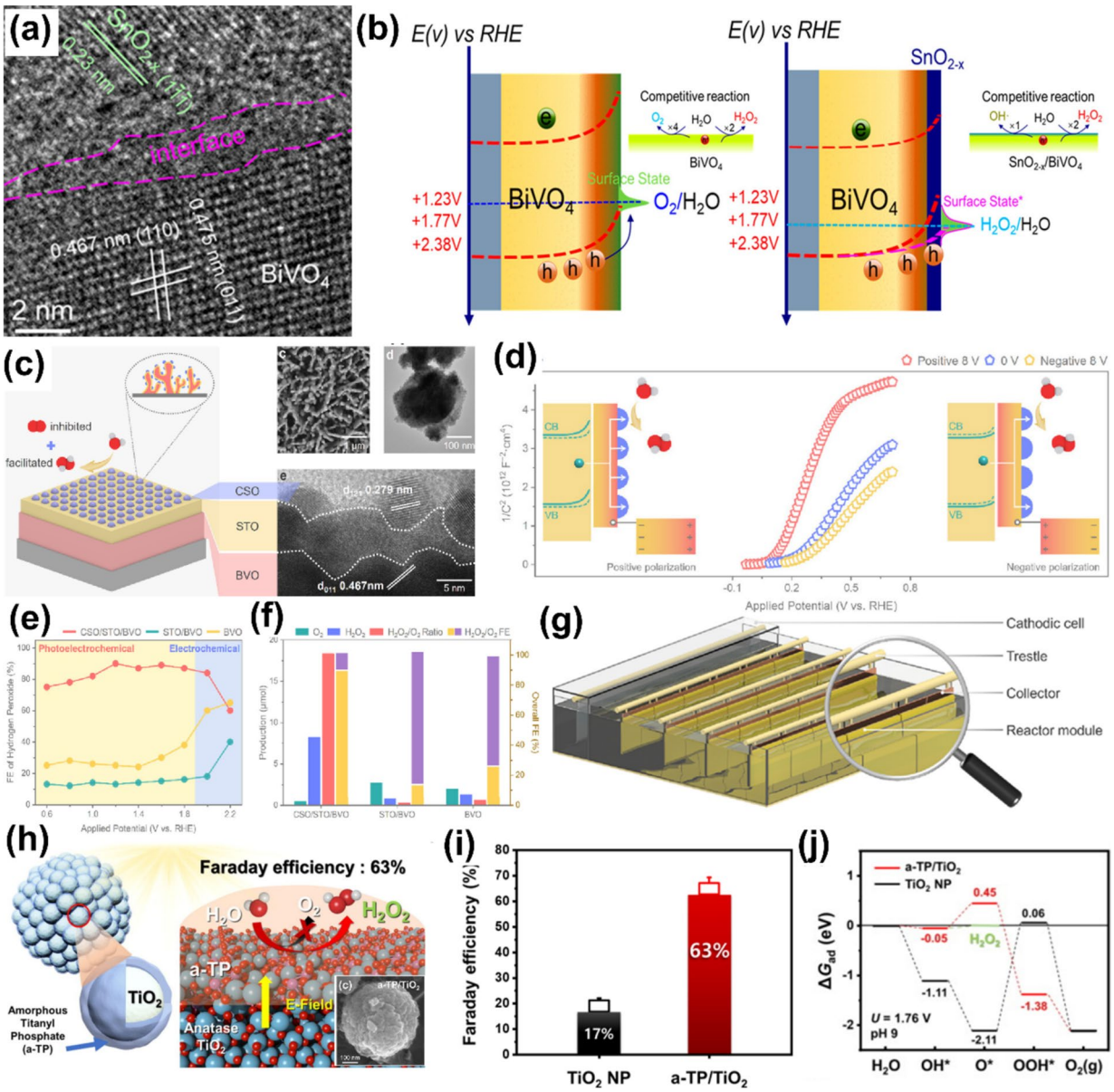

Fig. 7 **a** Morphology of $SnO_{2-x}/BiVO_4$. **b** Illustration of effect of SnO_{2-x} overlayer with the surface states which overcomes the surface OER. **c** Morphology of the CSO/STO/BVO photoanode. Adapted from [105]. Copyright 2020 American Chemical Society. **d** Schematic for carrier migration CSO/STO/BVO photoanode. **e–f** H_2O_2 production performance for CSO/STO/BVO photoanode. **g** Schematic diagram of CSO/STO/BVO photoanode in commercial application. Adapted from [121]. Copyright 2024 American Chemical Society. h Schematic illustration of a-TP/TiO_2 for PEC H_2O_2 production. **i and j** Faraday Efficiency and calculated energy diagrams of water oxidation for a-TP/TiO_2 in PEC H_2O_2 production. Adapted from [122]. Copyright 2023 Elsevier

nanoparticulate TiO_2 as substrate and enhanced its PEC H_2O_2 production capabilities significantly by decorating it with a titanyl phosphate cocatalyst (Fig. 7h). [122] The application of a titanyl phosphate cocatalyst overlayer on the TiO_2-based photoanode has led to improved conductivity and enhanced charge transfer efficiency. With titanyl phosphate decoration, the TiO_2 photoanode achieves an H_2O_2 production rate of approximately 13.50 μmol cm^{-2}, a figure nearly tenfold greater than the 1.32 μmol cm^{-2} observed for unmodified TiO_2. Additionally, the faradaic efficiency for the titanyl phosphate enhanced TiO_2 reached 63% at 1.76 V vs. RHE, showing a significant increase compared to the FE of the unadorned TiO_2. (Fig. 7i) [122]. The titanyl phosphate cocatalyst successfully neutralized the surface trap states on TiO_2, facilitating efficient charge transfer. Through DFT calculations, it was shown how the titanyl phosphate cocatalyst regulates the adsorption free energy of intermediates, steering the reaction along a controlled two-electron pathway that

favors high selectivity in PEC H_2O_2 generation. Furthermore, the incorporation of titanyl phosphate cocatalyst has amplified the intrinsic electric field within the TiO_2 photoanodes adorned with titanyl phosphate, which aids in enhancing the dynamics of hole transfer and transport (Fig. 7j).

Surface Regulation for PEC H_2O_2 Production

In PEC H_2O_2 production process, H_2O_2 is produced via the water oxidation reaction through the 2-electrons reaction pathway, which competes with the 4-electron-pathway-mediated OER. The product selectivity for the PEC WOR is also closely related to thermodynamic factors, including free energy of oxidative hole formation and energy of O_2 desorption. Nowadays, there are also some novel strategies that focusing on the above issue and achieved attractive improvements for PEC H_2O_2 production.

The free energy required for the formation of oxidative holes is influenced by the band gap and band-edge energies of photoanodes. As such, modifying the energy band structure of these materials represents a viable approach to altering the selectivity for H_2O_2 during PEC water oxidation processes. The integration of plasmonic nanostructures offers a promising method for enhancing the localized surface plasmon resonance effect within TMO semiconductors. This effect can capture and confine incoming electromagnetic waves, localizing the electric field near the semiconductor's surface by creating a potent near-field. Gyu Yong Jang et al. have developed a strategy for plasmon-driven band structure engineering aimed at thermodynamically optimizing product selectivity in PEC H_2O_2 production (Fig. 8a). [123] A nanopatterned array of plasmonic gold nanospheres (Au-PAT) was employed as the base layer beneath nanoparticulate $BiVO_4$ arrays. This arrangement led to a near-field coupling effect, generating a locally enhanced electromagnetic field around the $BiVO_4$. The influence of this plasmonic near-field was significant, altering the band structure of $BiVO_4$ and, consequently, the oxidation potential of holes within the H_2O_2 evolving photoanode. This modification resulted in an increased surface photovoltage of the photoanode and a reduction in band bending, indicating an enhanced potential difference for the quasi-Fermi level. (Fig. 8b). With the incorporation of the Au-PAT-driven plasmonic effect, the faradaic efficiency and production rate of H_2O_2 evolution were enhanced to 72.1% and 0.312 μmol cm^2 $min.^{-1}$, respectively. These figures represent an increase of 3.3 and 3.7 times over the baseline performance of unmodified $BiVO_4$. (Fig. 8c). [123]

Building hydrophobic microenvironment is another feasible choice for water oxidative H_2O_2 accumulation.[124] In the realm of PEC water oxidation reactions, a variety of approaches have been explored to enhance the activity of $BiVO_4$-based photoanodes, including surface passivation, heterojunction formation, heteroatom doping, and

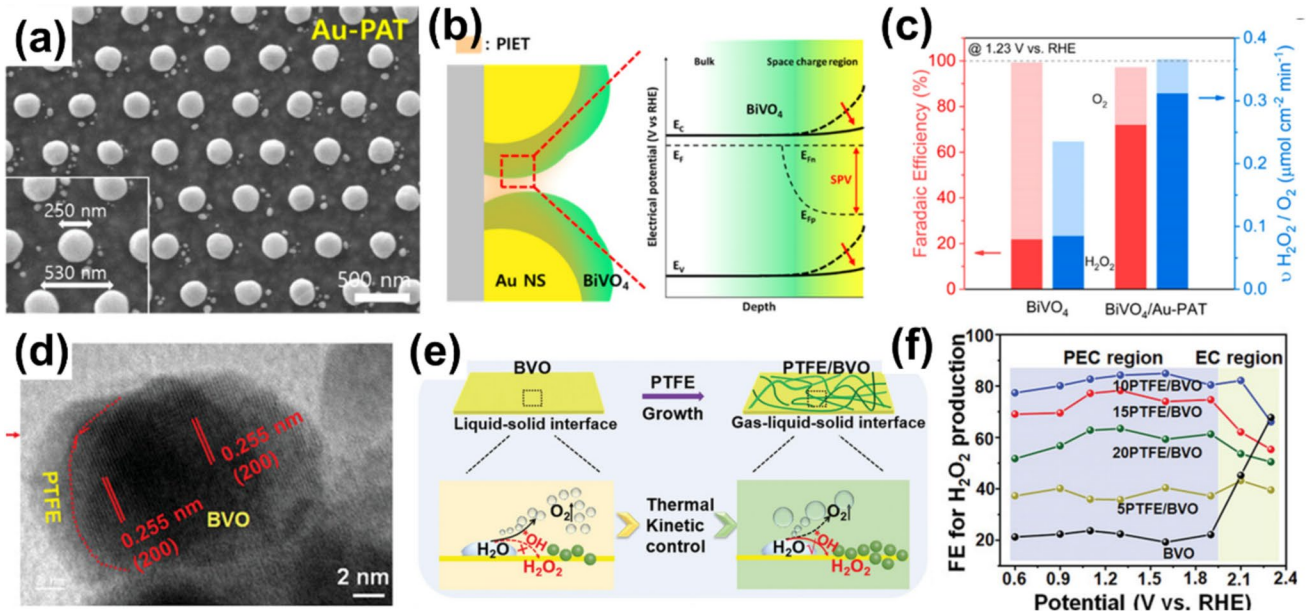

Fig. 8 **a** Morphology of Au-PAT array on the $BiVO_4$ substrate. **b** Schematic illustration of H_2O_2 production on the $BiVO_4$/Au-PAT. **c** Production yield and selectivity of the $BiVO_4$/Au-PAT photoanode for PEC H_2O_2 production. Adapted from [123]. Copyright 2023 American Chemical Society. **d** Structure PTFE/BVO photoanodes. **e** Schematic illustrations for regulating the ratios of O_2 gas and H_2O_2 liquid product on PTFE/BVO photoanode. **f** Faraday efficiency of the PEC H_2O_2 production of PTFE/BVO photoanode. Adapted from [126]. Copyright 2023 Wiley

facet control. These methods predominantly aim to modify the electronic properties of the $BiVO_4$ surface. However, the PEC water oxidation reaction inherently takes place at a hydrophilic bulk/electrolyte interface, leading to the in-situ decomposition of H_2O_2 prior to its desorption during prolonged operation [125]. Consequently, a hydrophobic gas–liquid-solid tri-phase interface is sought to improve the kinetics of H_2O_2 desorption, facilitating the separation of H_2O_2 before the reaction attains equilibrium. Thus, developing a suitably hydrophobic electrode surface emerges as a strategic approach to kinetically control the formation rate of O_2/H_2O_2, in turn boosting the selectivity and production rate of H_2O_2.

Man Ou et al. loaded polymer (PTFE) on $BiVO_4$ and modified a hydrophobic surface microenvironment. (Fig. 8e) [126] After incorporating PTFE, the average faradaic efficiency for H_2O_2 quadrupled to 81.6%, and the peak H_2O_2 concentration reached 150 μM at 1.23 V vs. RHE under AM 1.5 illumination over a continuous two-hour period (Fig. 8f). By applying a PTFE layer to create a hydrophobic surface, the water oxidation reaction (WOR) dynamics of the $BiVO_4$ photoanode are adjusted to thermodynamically prefer the two-electron pathway for H_2O_2 production, aided by the localized O_2 gas on the PTFE surface. Moreover, due to the hydrophobic nature of PTFE, H_2O_2 desorption in the aqueous solution is significantly enhanced and the decomposition of H_2O_2 is effectively reduced in the kinetics. The newly developed PTFE/$BiVO_4$ photoanode also exhibits improved current density stability compared to the original $BiVO_4$. [126]

Summary of PEC H_2O_2 Production

Replacing anodic O_2 generation with H_2O_2 could improve the economic attractiveness of PEC water splitting. Here, Wenderich et. al. presented a techno-economic analysis of a PEC H_2/H_2O_2 process, and then simulated a PEC farm with industrially relevant production capacity. They pointed out that significant reduction in the levelized cost of hydrogen can be realized within the PEC community when future research is aimed at enhancing the stability of the photoanode and optimizing the solar-to-hydrogen efficiency for anodic H_2O_2 formation [127]. Hou's laboratory fabricated e large area $BiVO_4$-based tandem cell of 3×3 cm^2 can reach a total H_2O_2 production yield of 338.84 μmol L^{-1}, which paves the way for the rational design and fabrication of artificial photosynthetic cells for the production of liquid solar fuel. [128]. Kan Zhang et al. designed a 144 cm^2 array panel reactor, which achieve an industrial-scale peak photocurrent of 0.11 A under a natural irradiance of 862 $W \cdot m^{-2}$, enable further optimization of industrial applications[129].

A summary table is given for recent progress of PEC H_2O_2 production as Table 2. Up to now, the products detected from PEC water splitting without an OER catalyst have been mostly H_2 and O_2, but the Faraday efficiency for O_2 production is usually lower than 100%. Few studies have investigated whether H_2O_2 is produced in the water oxidation process, and the mechanism of H_2O_2 production remains unclear, especially regarding the regulation of product selectivity. Further research into the determinants and general principles governing the selectivity and activity of photocatalytic water oxidation products holds significant reference value for adjusting the photocatalytic water oxidation process. Additionally, catalytic reactions are usually thermodynamically controlled. Most strategies aimed at achieving

Table 2 Progress of value-added solar-to-hydrogen energy conversions via PEC H_2O_2 production

Material	Electrolyte	Current Density (mA cm^{-2})	Stability (h)	Applied Bias (V vs. RHE)	Faradic Efficiency (%)	Ref
$BiVO_4$	1 M $KHCO_3$	3.3	1	1.2	40	[105]
SnO_2/ $BiVO_4$	1 M $KHCO_3$	4	1	1.2	86	[105]
CSO/STO/BVO	1 M $NaHCO_3$	3.89	1	1.2	85	[121]
$BiVO_4$/Au-PAT	1 M $NaHCO_3$	1.66	2	1.23	72.1	[123]
Gd-doped $BiVO_4$	2 M $KHCO_3$	50	3	3	44	[150]
WO_3-$BiVO_4$	2 M $KHCO_3$	4	5	1.5	54	[115]
WO_3-Mo:$BiVO_4$	2 M $KHCO_3$	5	6	0.8	68	[151]
P-Mo:$BiVO_4$	1 M $KHCO_3$	1.25	3	1	50	[117]
A-TP/TiO_2	1 M $KHCO_3$	0.46	3	1.76	63	[122]
Co_3O_4/TiO_2	0.5 M $KHCO_3$	0.78	-	1.77	27	[152]
$NiTiO_3$	0.1 M K_2HPO_4	2	3	1.23	76	[153]

high reaction activity and high selectivity of target products focus on lowering the reaction energy barrier or increasing the speed of key reaction steps. However, for reactions that are challenging to control thermodynamically or have few intermediates, there is still a lack of effective strategies for achieving high reactivity and product selectivity. Therefore, future research on PEC H_2O_2 production should develop strategies for comprehensive tripartite coordination of photo-carrier kinetics, water oxidation kinetics, and H_2O_2 desorption kinetics to regulate photocatalytic activity and product selectivity.

Photoelectrochemical Ammonia Splitting

The numerous complexities involved in handling hydrogen fuel have prompted many to consider the ammonia economy. This section discusses the PEC AOR as a means of solar-to-chemical energy conversion, which is for on-site hydrogen production via PEC decomposition of ammonia as a fuel. The development of robust and stable PEC AOR catalysts is critical for both applications. This section explains the barriers that need to be overcome in catalyst development, and clarifies the kinetics and mechanism of PEC AOR, laying the foundation for a better understanding of the issues associated with this technology, and comprehensively reviews the Pt-free catalysts that have been explored for AOR to promote future research directions for this promising technology.

Basic Principles of PEC AOR

The oxidation potential of NH_3 is 0.06 V_{RHE}, which is significantly lower than that of water oxidation reaction (1.23 V_{RHE}). The ammonia oxidation reaction usually occur in alkaline solution, the photoanode and photocathode reactions are presented as follows: [130]

$$\text{Anode}: 2NH_3 + 6OH^- \rightarrow N_2 + 6H_2O + 6e^- \quad E = -0.83Vvs.RHE \tag{5}$$

$$\text{Cathode}: 6H_2O + 6e^- \rightarrow 3H_2 + 6OH^- \quad E = -0.77Vvs.RHE \tag{6}$$

Theoretically, employing ammonia as the reactant in PEC hydrogen production can significantly reduce the applied bias at the photoanode, facilitating a 95% reduction in the energy required for hydrogen production[131]. Moreover, ammonia is a major substance causing water pollution, which can cause toxic effects on humans and other living organisms[132]. In recent years, with the growing public expectation for higher quality living environments and the gradual increase in environmental protection standards, the removal of ammoniacal nitrogen compounds from wastewater has become a crucial aspect of environmental pollution management [133]. By switching the anodic reaction from water oxidation to ammonia oxidation, it is possible to directly utilize solar energy to decompose ammonia in water into N_2 at anode, while H_2 can be generated at the cathode [134]. Thus, it becomes possible to obtain hydrogen as a green and environmentally friendly fuel while simultaneously degrading ammoniacal nitrogen pollutants in water.

Although the PEC ammonia splitting offers a higher solar to hydrogen conversion efficiency compared to the conventional PEC water splitting process, this technology still faces some challenges. Additionally, intermediates such as $*NO_x$ and $*N_2$ functional groups produced during the ammonia oxidation reaction are difficult to desorb from the surface of the photoanodes, leading to catalyst deactivation. This results in poor long-term stability of the photoanodes and a short operational durability [135, 136].

To render the PEC ammonia splitting technology practical, it is imperative to enhance the selectivity of the photoanode materials towards the products of the ammonia oxidation reaction, ensuring the complete oxidation from NH_3 to N_2. Additionally, it is necessary to suppress the formation of $*NO_x$ and $*N_2$ to improve the long-term stability of the photoanode materials. Currently, there are two main conjectures regarding the pathways of the ammonia oxidation reaction, as shown in Fig. 9.[137] The first is the O-S mechanism, where the amine group continuously dehydrogenates into nitrogen atoms, followed by the formation of N≡N bonds to produce N_2; the second is the G-M mechanism, where the amine group partially dehydrogenates into the dehydrogenated intermediate $*NH_x$, which then dimerizes into hydrazine substances $*N_2H_x$, and finally dehydrogenates to form N_2. [137] In both conjectured ammonia oxidation processes, multiple stepwise reaction steps are required to fully decompose NH_3 into N_2 and H_2. Therefore, to achieve photoanode materials for PEC ammonia splitting with high stability and selectivity for the desired products, the following strategy must be implemented: modulate the reaction kinetics of the photoanode materials to facilitate reactions between photogenerated holes and specific surface adsorbates (including groups, molecules or ions) [138]. This will suppress the formation of $*NO_x$ and reduce the desorption activation energy of $*N_2$, enabling the complete oxidative decomposition of NH_3 on the photoanode surface [139].

Research Frontier of PEC Ammonia Splitting

Research on PEC ammonia splitting is currently at a nascent stage with limited findings, primarily focusing on the technology's feasibility and fundamental reaction mechanisms.

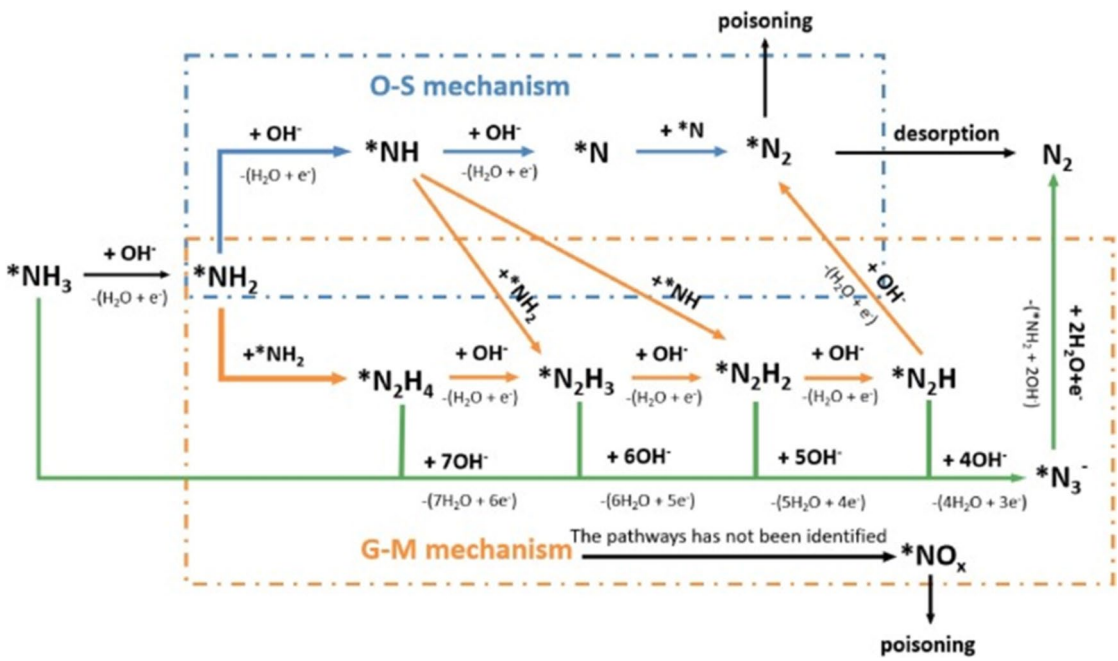

Fig. 9 Illustration of ammonia oxidation reaction with multiple products and intermediates. Adapted from [137]. Copyright 2022 Elsevier

Hence, there is a broad scope for further research in PEC ammonia splitting.

In 2021, Jung Kyu Kim et al. successfully conducted the PEC ammonia decomposition in aqueous solutions with iron phosphate decorated Fe_2O_3 ($FePi/Fe_2O_3$) as photoanode materials (Fig. 10a). For the first time, it demonstrated that decomposing NH_3 at 1.23 V vs. RHE with solar assistance can be a more efficient strategy rather than photocatalytic and electrocatalytic strategies. In the simulated ammonia wastewater electrolyte with is 1 M NaOH solution and 500 ppm NH_3, by applying the PEC ammonia decomposition treatment with 1 sun illumination and 1.23 V vs. RHE bias, about 54.4% of the was decomposed after 3 h. However, in the electrocatalytic and photocatalytic processes, after 3 h, the decomposition rate of ammonia was only about 22.6% and 32.0%, respectively. (Fig. 10b) The research posited that the adsorption of NH_3 molecules on the surface plays a crucial role in the ammonia oxidation reaction. Under solar illumination, electron/hole pairs are produced within the bulk of $\alpha-Fe_2O_3$.With the surface FePi undergoing continuous oxidation, the recombination of photogenerated electron/hole pairs are suppressed, thereby facilitating hole transfer and boosting NH_3 molecule adsorption. Consequently, in the PEC ammonia oxidation reaction, the applied bias required is significantly lower compared to the electrocatalytic ammoxidation process, demonstrating enhanced catalytic efficiency at 1.23 V vs. RHE. (Fig. 10c). [140].

Lei Wu et al. undertook a comprehensive analysis of the step-by-step reaction mechanism in the PEC AOR utilizing Fe_2O_3 photoanodes [141]. They detected the compositions of anode products systemically, and proved the existence of the AOR products (N_2, NO_2^- and NO_3^-) and the WOR product O_2 (Fig. 10d), along with small amounts of hydrazine (N_2H_4) and trace amount of NO. Regarding the competition between AOR and WOR at the photoanode, due to NH_3 participating in the surface hole-capturing mechanism, NH_3-related surface states, like $Fe^{III}-NH_2$ are expected to form, alongside surface states related to H_2O/OH^-. (Fig. 10e) Hence, they proposed a non-radical nucleophilic attack mechanism for PEC AOR. The selectivity of the AOR can be finely adjusted by altering the balance between two types of surface species on α-Fe_2O_3: $Fe-O$ and $Fe-N$. In the presence of NH_3, $Fe^{IV}=O$ is more likely to engage with the stronger nucleophile NH_3, leading to the formation of $N-O$ bonds and the subsequent production of NO_2^-. NO_3^- can then be produced via oxygen atom transfer from $Fe^{IV}=O$ to NO_2^-. Conversely, $FeIII-NH_2$ results from the exchange reaction between $Fe^{III}-OH$ and NH_3. Photogenerated holes oxidize $Fe^{III}-NH_2$ to create high-valency $Fe-N$ surface species. NH_3's subsequent nucleophilic attack on $Fe^{IV}=NH$ sites generate the $Fe^{III}-N_2H_y$ intermediate. This intermediate either directly desorbs from the $\alpha-Fe_2O_3$ surface as N_2H_4 or undergoes further dehydrogenation to release N_2.[141] Furthermore, Jincai Zhao's research group experimented with incorporating trace amounts of Cu ions (ranging from 0.5 to 10 mg L^{-1}) into NH_3 electrolyte, which markedly enhanced the AOR photocurrent in unmodified $BiVO_4$ photoanodes. [142] The AOR demonstrates a high Faradaic efficiency of 93.8%, with the WOR significantly diminished. The enhancement of AOR on the $BiVO_4$ photoanode is attributed

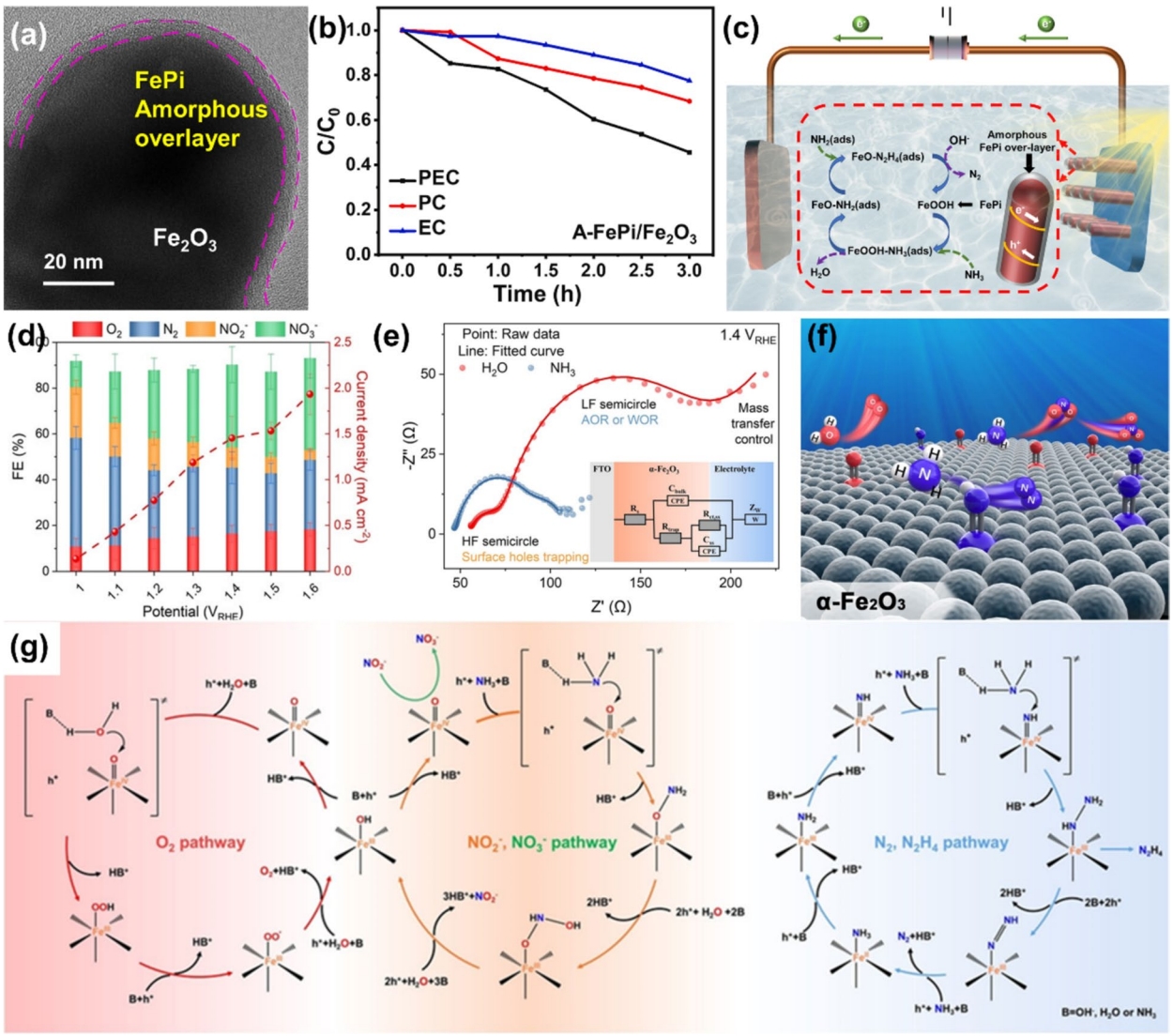

Fig. 10 **a** Morphology of FePi/Fe_2O_3 photoanode for PEC AOR. **b** Ammonia Degradation rate of activated FePi/Fe_2O_3 photoanode in PEC, electrocatalytic and photocatalytic condition. **c** Illustration of FePi/Fe_2O_3 photoanode for PEC AOR. Adapted from [140] Copyright 2020 Elsevier. **d** Faraday efficiency for AOR and WOR at different potentials on α-Fe_2O_3 photoanodes. **e** Charge migration characteristic in AOR. **f and g** Illustration of α-Fe_2O_3 photoanodes in PEC AOR and the competitive non-radical nucleophilic attack pathways for ammonia oxidation. Adapted from [141]. Copyright 2022 Wiley

to the improved NH3 adsorption, facilitated by the presence of Cu ions. The Cu – NH_3 complexes show a preferential adsorption on $BiVO_4$ surfaces, effectively outcompeting WOR. Furthermore, the synergy between the surface effect of $BiVO_4$ and the Cu-induced coordination effect plays a crucial role in activating N – H bonds, thus speeding up the initial, rate-limiting step of proton-coupled electron transfer for AOR. This study underscores the importance of modulating the NH_3 adsorption process to govern the competitive dynamics between AOR and WOR [142].

At present, research on PEC AOR technology is mainly focused on the system of NH_4^+/NH_3 oxidation, while the industrial system for PEC AOR is still in its infancy, and more attention should be paid to the development of various environmental materials, especially nanostructured materials. Although PEC AOR can effectively degrade NH_3 in laboratory-scale experiments, the actual application of PEC AOR in the treatment of ammonia-containing wastewater is still in progress. For potential applications, the design and optimization of reactor systems, actual condition data, pilot and full-scale experiments, and economic evaluation deserve more research.

Conclusion and Perspective

Value-added reactions utilize highly efficiency reaction process, such as GOR, $2e^-$ WOR, AOR, to produce green H_2 fuel coupled with high profits, which is a promising strategy for green solar-to-chemical energy conversions. Among the solar-to-chemical energy conversion technologies, the electricity/solar driven redox reaction have been considered to be appealing choice. Electrocatalysis (EC) is one of the most widely used energy conversion technologies and has the advantages of high efficiency, high adaptability, and easy automation control, but the huge electrical consumption and exorbitant cost need to be solved. Photocatalysis (PC) is considered to be a water splitting technique with great potential due to its environmentally friendly and low energy-consuming. Nevertheless, the degradation performance of PC is limited by the fast recombination of photogenerated electron/hole pairs. Additionally, the difficult separation and recovery of the catalyst may lead to serious pollution. Compared with EC and PC processes, PEC is a process driven by PC and EC simultaneously, a bias is introduced to cause the photogenerated electrons to flow to the counter electrode through an external circuit, resulting in the enhanced separation of the electrons from the holes, which can significantly improve the efficiency of reactive species generation in PC and conducive to the conversion of solar energy to chemical energy.

In this review, we summarize current progress in developing economically friendly PEC reactions for value-added solar-to-chemical energy conversions, especially for glycerol oxidation, H_2O_2 production and ammonia splitting. Replacing the sluggish water oxidation reaction with the above value-added reaction can achieve better financial return, while also generating green hydrogen fuel with higher energy conversion efficiency. Such attractive advantages make designing PEC value-added upgrading energy conversions becoming a promising strategy for comprehensive utilizing of solar energy. However, there are still many challenges in obtaining desirable products with high selectivity and yield that need to be overcome in the near future.

Firstly, the mechanism of PEC value-added upgrading reactions should be studied in-depth. For example, in the case of glycerol oxidation, different products are produced when different hydroxyl groups serve as the active sites of the reaction, and thorough analysis of the reaction intermediates and final-products generated during the reaction is important for achieving higher selectivity of specific products. Similar challenges also exist in PEC H_2O_2 production and ammonia splitting. Therefore, in-depth studies are needed to understand each reaction pathway and mechanism of different reactants, which will help continue to advance catalyst design and the development of reaction systems.

Secondly, all of these three PEC value-added upgrading reaction are measured in aqueous phase, which are heterogeneous catalytic process. Since competition with water oxidation reactions is inevitable, reaction kinetics become extremely complex and often depend in part on the concentration of reactants. Fully upgrading the reactant into a desirable product is difficult to achieve. A commitment to clever reaction system design is required, and kinetic regulation mechanism of heterogeneous competitive reactions should be systemically researched.

Thirdly, rational designing and optimizing large-scale PEC cells with cyclic streamlines is essential for developing PEC value-added energy conversion system. Light absorption and products separation are important issues when designing PEC energy conversion system. With the existing of glycerol, ammonia, or other reactants, viscosity of electrolyte might be different to that of water oxidation reactants, which may exhibit slower mass transfer, resulting in a reduced reaction activity of the photoanode. Nowadays, most laboratory-scale PEC reactions are performed using single or H-cells, but execution remains challenging due to the difficulty to adjust productivity and effectively separate products. Therefore, it is necessary to systematically design large-scale PEC cells with the engineering sight and establish PEC value-added energy conversion systems to obtain higher profits.

In general, by regulating the reaction pathway of PEC value-added energy conversion reaction, especially anode reaction kinetics to promote direct oxidation of reactants and indirect oxidation of various intermediates, can be applied to produce different high-value products. Therefore, PEC value-added energy conversions can not only couple hydrogen production while upgrading small organic molecules to high value-added chemicals, but also provide opportunities for biomass reforming and pollutant degradation. This investigation on PEC value-added reactions holds promising potential for attaining economic benefits and exhibits application prospects in advancing PEC energy conversions.

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Data Availability This review paper does not include experimental data generated by the authors. Thus, data sharing is not applicable.

Declarations

Conflict of interest The authors declare no conflict of interest.

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